

Question	Marking Guidance	Mark	Comments
9.1	- Tracheoles have thin walls so short diffusion distance to cells; - Highly branched/large number of tracheoles so short diffusion distance to cells; - Highly branched/large number of tracheoles so large surface area (for gas exchange); - Tracheae provide tubes full of air so fast diffusion (into insect tissues); - Fluid in the end of the tracheoles that moves out (into tissues) during exercise so faster diffusion through the air to the gas exchange surface; OR Fluid in the end of the tracheoles that moves out (into tissues) during exercise so larger surface area (for gas exchange); - Body can be moved (by muscles) to move air so maintains diffusion/concentration gradient for oxygen/carbon dioxide;	3 max	1. Do not accept unqualified references to thin membranes. Max 2 if any reference to blood Ignore references to spiracles 5. Accept 'water' for fluid. Accept 'cells' and 'tissues' as interchangeable words.
9.2	- Damselfly larvae has high(er) metabolic/respiratory (rate); (So) uses more oxygen (per unit time/per unit mass);	2	Idea of 'more/high' is needed for both mark points. 2. Accept 'needs' for 'uses' 2. Ignore references to absorbing/obtaining/uptake of more oxygen
9.3	Mean SA = $9.85 \text{ mm}^2/9.9 \text{ mm}^2$;; Percentage uncertainty of SA = $18.5/18.7/19$;; If both answers incorrect 1 mark for Percentage uncertainty of dimensions $11.8/12$ and $6.70/6.7$ Surface area correctly calculated with correct units but not rounded to appropriate sf (9.8532 mm^2) Surface area correct (with appropriate sf) but	3 max	Both answers correct = 3 marks 1 answer correct only = 2 Both answers incorrect = max 1

	no/incorrect unit given		
9.4	1. Don't use shading; 2. Only use single lines/don't use sketching (lines)/ensure lines are continuous/connected; 3. Add further labels/annotations; 4. Don't cross label lines; 5. Add magnification/scale (bar);	2 max	Reject 'colour in'. Reject 'use of electron microscopes' Ignore 'use a sharp pencil'